

CCSU BCA 1st Semester – Problem Solving Techniques

2026 Exam-Oriented Important Notes

Unit I – Problem Solving Basics

- Problem definition and analysis
- Problem-solving steps
- Algorithm and characteristics
- Flowchart symbols and construction
- Pseudocode
- Algorithm vs flowchart

Unit II – Programming Fundamentals

- Variables and constants
- Data types
- Input/output
- Arithmetic, relational and logical operators
- Conditional statements: if, if-else, switch
- Loops: for, while, do-while

Unit III – Structured Programming

- Functions and parameter passing
- Recursion
- Arrays: 1-D and 2-D
- Strings and basic string operations
- Searching and sorting basics
- Debugging and testing

Unit IV – C Programming Practice

- Basic C program structure
- Programs: even/odd, largest of numbers, factorial, prime, palindrome, Fibonacci
- Functions with arrays
- Basic file/program error concepts
- Dry run and tracing of programs

■ Important Questions for Revision

- Define algorithm and explain its characteristics.
- Draw/describe common flowchart symbols.
- Compare if-else and switch.
- Explain for, while and do-while loops.
- Write programs for factorial, prime, palindrome and Fibonacci.
- Explain functions, arrays and recursion.

Note: These are revision-oriented notes aligned to the current CCSU BCA 2025–26 subject structure; they are not a prediction of the exact 2026 paper.